

Architecture Overview

Solution Architecture

The application was designed as a simple FileMaker solution focused on clarity and maintainability. The data model is built around four core entities: Products, Customers, Orders, and OrderItems.

The current implementation is intended to demonstrate the core business process of creating and managing orders while keeping the frontend simple. Business rules are implemented through FileMaker scripts and calculations, making them easy to understand and modify.

In a production environment, the solution would be hosted on FileMaker Server, allowing multiple users to work with the system simultaneously. If required, external systems such as a webshop, ERP, or reporting platform could connect through the FileMaker Data API.

Scalability

The solution separates master data (Products and Customers) from transactional data (Orders and OrderItems), which helps keep the structure organized as the system grows.

As the number of records increases, performance can be maintained by using indexed fields, efficient relationships, and server-side processing for more demanding tasks. The FileMaker Data API could also be used to receive orders from an external webshop without changing the core application structure.

The current design provides a solid foundation that can be expanded without major changes to the data model.

Potential Bottleneck

The most likely bottleneck is stock handling during order completion.

In a larger multi-user environment, multiple users could attempt to complete orders containing the same product at the same time. This creates a risk of record locking and stock conflicts.

For simplicity, stock is currently stored directly on the product record. In a larger solution, I would introduce an inventory movement model where each stock change is recorded as a transaction. This would provide a complete audit trail and improve handling of concurrent stock updates.

If the application needed to support a larger number of users or higher transaction volumes, I would move stock management to a dedicated inventory movement model or server-side process. This would provide better auditability and reduce the risk of conflicts when multiple users work with the same products simultaneously.